

# YASMEEN OUF ABDELRAHMAN

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## RESEARCH PROFILE

Academic researcher specialising in deep learning for medical image analysis, with a focus on hybrid CNN-Transformer architectures, multimodal learning, and explainable AI (XAI) for clinical diagnostics. MSc (Distinction) from the University of Greenwich; active Visiting Fellow and research collaborator within the Faculty of Engineering & Science. Published in IEEE with work spanning medical imaging, neurological disease classification, and sustainable computing. Five years of undergraduate teaching experience across UK and Egyptian universities. Seeking a fully funded PhD to advance research at the intersection of AI, computer vision, and biomedical informatics.

## RESEARCH EXPERIENCE

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| <b>Visiting Fellow (Remote)</b><br><i>University of Greenwich, Faculty of Engineering &amp; Science</i>                                                                                                                                                                                                                                                                                                                                                                                                                                    | 2025 – Present<br>London, UK |
| <ul style="list-style-type: none"><li>Conducting collaborative research with Dr K. Kapinchev on hybrid CNN-Transformer architectures for mammographic mass classification, resulting in an accepted IEEE conference paper (AMLDs 2026).</li><li>Actively contributing to the faculty's medical imaging research agenda through literature reviews, experimental design, and manuscript preparation.</li><li>Participating in departmental research seminars and supporting ongoing grant-funded projects.</li></ul>                        |                              |
| <b>Research-Active Assistant Lecturer</b><br><i>MSA University</i>                                                                                                                                                                                                                                                                                                                                                                                                                                                                         | 2022 – 2025<br>Giza, Egypt   |
| <ul style="list-style-type: none"><li>Supervised undergraduate graduation projects, one of which directly resulted in a peer-reviewed IEEE Xplore publication (Multiple Sclerosis Classification and Detection, IMSA 2024).</li><li>Co-supervised student research teams on machine learning and computer vision projects, guiding experimental methodology, dataset selection, and evaluation.</li><li>Served on the faculty graduation project reviewing committee, conducting technical assessment of documentation and code.</li></ul> |                              |

## EDUCATION

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| <b>MSc Computer Science — Distinction</b><br><i>University of Greenwich</i>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   | 2021 – 2022<br>London, UK  |
| <ul style="list-style-type: none"><li>Thesis: Deep Learning Model for Accurate Detection of Recyclable Materials</li><li>Developed a deep learning classification pipeline achieving high accuracy on recyclable material categories, contributing to automated waste-sorting research.</li><li>Relevant modules: Machine Learning, Computer Vision, Research Methods in Computing, Advanced Databases, Software Engineering.</li><li>Awarded fully funded MSc scholarship by MSA University (2020); selected competitively from graduating cohort.</li></ul> |                            |
| <b>BSc Computer Science — GPA 3.2</b><br><i>MSA University (Validated by University of Greenwich)</i>                                                                                                                                                                                                                                                                                                                                                                                                                                                         | 2015 – 2019<br>Giza, Egypt |
| <ul style="list-style-type: none"><li>Graduation Project: Detection of Cervical Cancer Using Computational Genome Assembly</li><li>Designed a bioinformatics pipeline for early cervical cancer detection using Next-Generation Sequencing (NGS) data.</li><li>Implemented genome assembly techniques and Burrows-Wheeler Transform (BWT) algorithms for sequence alignment.</li><li>Relevant modules: Data Structures &amp; Algorithms, Bioinformatics, Pattern Recognition, Software Development.</li></ul>                                                 |                            |

## TEACHING & ACADEMIC EXPERIENCE

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### Assistant Lecturer

MSA University

2022 – 2025

Giza, Egypt

- Designed and delivered lectures, tutorials, and workshops for undergraduate Computer Science courses including Machine Learning, Programming, and Software Engineering.
- Set, administered, and graded assessments, maintaining alignment with programme learning outcomes.
- Worked within the examination control unit and supported the student registration process.

### Teaching Assistant

University of Greenwich

2021 – 2022

London, UK

- Supported students during tutorials and drop-in sessions; clarified course concepts across programming and computer science modules.
- Collaborated with academic staff to improve the digital learning experience for the course cohort.

### Teaching Assistant

MSA University

2019 – 2021

Giza, Egypt

Assisted in teaching and grading undergraduate Computer Science courses; provided academic guidance and mentorship to students.

## RESEARCH & TECHNICAL PROJECTS

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### Hybrid CNN-Transformer for Mammographic Mass Classification

- Designed a multimodal deep learning architecture combining DenseNet-169 feature extraction with a DeiT-Tiny Vision Transformer and coordinate attention for breast mass classification across INbreast, MIAS, and CBIS-DDSM datasets.
- Integrated GradCAM-based explainability to generate clinically interpretable saliency maps, supporting XAI validation methodology.
- Results accepted for publication at AMLDS 2026 (IEEE).

### LSTM vs BiLSTM for Short-Term Soil Moisture Forecasting

- Built and compared LSTM and BiLSTM forecasting models in PyTorch for predicting soil moisture 30 minutes to 6 hours ahead using 10-minute resolution data from the FESSTVaL campaign (Falkenberg, Germany).
- Designed a fully reproducible end-to-end pipeline; demonstrated BiLSTM networks more effectively capture bidirectional temporal patterns in hydrological time-series.

### Machine Learning for Facial Attribute Classification

Implemented and benchmarked CNN and VGG16 architectures for multi-label facial attribute prediction; evaluated trade-offs in accuracy, inference speed, and generalisation across demographic subgroups.

### Customer Insights Data Analysis

Analysed transactional retail data using advanced visualisation (heatmaps, scatter plots, clustering); identified actionable patterns in customer behaviour and operational inefficiencies.

## PUBLICATIONS

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**Abdelrahman, Y.A.O.H.** & Kapinchev, K. (Accepted, 2026). Hybrid CNN-Transformer Architecture for Mass Classification. *2026 International Conference on Advanced Machine Learning and Data Science (AMLDS 2026)*, Osaka, Japan. IEEE. [In Press — IEEE Xplore pending]

Sharabas, Y.H.M.F., **Abdelrahman, Y.A.O.H.**, & Taha, Z.A.E.H. (2024). Multiple Sclerosis (MS) Classification and Detection. *Intelligent Methods, Systems, and Applications (IMSA 2024)*. IEEE Xplore. DOI: 10.1109/IMSA61967.2024.10652740

## TECHNICAL SKILLS

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| <b>Languages</b><br>Python, Java, JavaScript, SQL, R              | <b>Data &amp; Tools</b><br>pandas, NumPy, Matplotlib, Seaborn, FAISS |
| <b>ML / DL</b><br>PyTorch, TensorFlow, Keras, scikit-learn        | <b>Dev Tools</b><br>Git/GitHub, Docker, Jupyter, LaTeX               |
| <b>Computer Vision</b><br>CNN, ViT, DenseNet, DeiT, GradCAM, YOLO | <b>Databases</b><br>MySQL, SQL Server, PostgreSQL                    |
| <b>NLP / XAI</b><br>Transformers, LIME, SHAP                      | <b>Platforms</b><br>Google Colab, AWS (basic), Linux/Ubuntu          |

## AWARDS & FUNDING

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**Fully Funded MSc Scholarship**  
*MSA University / University of Greenwich*  
Competitive scholarship covering full tuition and living costs for the MSc in Computer Science at the University of Greenwich, Faculty of Engineering & Science.

## LANGUAGES

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|------------------------------------------------------------------------------|---------------------------------|
| <b>English</b><br>Excellent — written and spoken (academic and professional) | <b>Arabic</b><br>Native speaker |
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## REFERENCES

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References available upon request